

ROHIT KUMAR

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Skills

Languages: Python, C++, SQL, JavaScript

Machine Learning: Supervised Learning, Unsupervised Learning, Deep Learning

Libraries: NumPy, Pandas, Seaborn, Matplotlib, Scikit-learn

Web Development: HTML, CSS, Bootstrap, MySQL, MongoDB

Core CS: DSA, OOP, DBMS, Operating Systems

Problem Solving: Solved 100+ DSA problems across multiple competitive coding platforms

Tools: Git/GitHub, Linux, Anaconda, Jupyter Notebook, VS Code

Research Experience

M.E. Thesis – Entropy-Guided CNN for Cocaine Use Disorder Detection (MRI)

2024 – 2026

Thapar Institute of Engineering and Technology

- Developed entropy-based slice selection and CNN framework for structural MRI classification.
- Processed and analyzed 142-subject MRI real-world dataset SUDMEX-CONN using automated preprocessing and leakage-free evaluation.
- Built preprocessing pipeline using FSL (skull stripping, normalization, MNI registration).
- Achieved 72% subject-level accuracy (AUC 0.69) with leakage-free evaluation.
- Leveraged SHAP to explain CNN predictions and identify key brain regions, improving clinical interpretability.

Projects

CataractAI – Deep Learning Cataract Prediction System | GitHub

- Developed computer vision-based medical AI system to classify cataract severity from retinal fundus images.
- Achieved 90%+ validation accuracy using EfficientNet-0 with transfer learning.
- Designed end-to-end ML pipeline for preprocessing, training, and inference.
- Developed responsive web interface for real-time image upload and diagnosis.
- Technologies: Python, TensorFlow, MySQL

Hotel Management System – Full Stack Web Application | GitHub

- Developed a full-stack hotel operations platform to manage rooms, bookings, customers, and billing workflows.
- Implemented MySQL database schema for rooms, reservations, payments, and invoices.
- Built responsive dashboard with real-time analytics, booking management, and revenue reports.
- Deployed modular frontend with HTML5, CSS3, Bootstrap, and dynamic JavaScript components.
- Technologies: Express.js, MySQL, HTML, CSS, JavaScript

Internship Experience

Frontend Developer Intern

Jan 2022 – May 2022

PN Infosys

- Developed 10+ responsive webpages using HTML5, CSS3, JavaScript, and Bootstrap, improving mobile usability and layout consistency.
- Optimized UI performance and reduced page load time by 20% through code refactoring and asset compression.
- Fixed 25+ UI bugs and improved cross-browser compatibility.
- Collaborated with backend team to integrate REST APIs and dynamic features.

Education

M.Tech – Computer Science

2024 – 2026

Thapar Institute of Engineering and Technology

CGPA: 7.46

B.Tech – Information Technology

2019 – 2023

Rustamji Institute of Technology

CGPA: 8.36